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**Investigations of Hot and Cold Nuclear Matter Effects in Small Systems with
the ATLAS Detector**

by

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Dissertation

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Abstract

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Acknowledgements

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A dedication is optional.

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List of Abbreviations

Abbreviation	Definition
ATLAS	A Toroidal LHC ApparatuS
LHC	Large Hadron Collider
QCD	Quantum Chromodynamics
QGP	Quark-Gluon Plasma
ZDC	Zero Degree Calorimeter

List of Symbols

Symbol	Definition
$\sqrt{s_{\text{NN}}}$	Nucleon-nucleon center-of-mass energy
p_{T}	Transverse momentum
η	Pseudorapidity
ϕ	Azimuthal angle

Chapter 1 Introduction

The goal of physics is to create a mathematical model that accurately describes a physical system. Different types of theories may be appropriate for describing different energy and length scales. Interactions between relatively large and slow objects, all of those we encounter in day-to-day life, fall under the purview of Classical Mechanics. When these large objects are accelerated to speeds approaching the speed of light, Relativistic Mechanics describes their interactions well. When one instead views the world at the smallest scales, one finds that the universe is made up of a few types of fundamental particles which are identical and indistinguishable. The behavior of this fundamental matter at low energies is given by Quantum Mechanics, while at high energies it is described by Quantum Field Theory. Quantum Field Theory (QFT) represents the combination of relativity and quantum mechanics.

add bridge to standard model

1.1 The Standard Model

The Standard Model of particle physics represents the most complete description of fundamental interactions to date, surviving over 70+ years of theoretical and experimental scrutiny. The Standard Model, shown in Figure 1, combines Quantum Field Theories of the ElectroWeak forces (Electrodynamics) and the Strong interaction (Chromodynamics) into a unified framework. Each interaction has its own force carrier, with the strong, electromagnetic, and weak forces being mediated by the gluon, photon, and $W^\pm + Z$ bosons respectively.

Standard Model of Elementary Particles

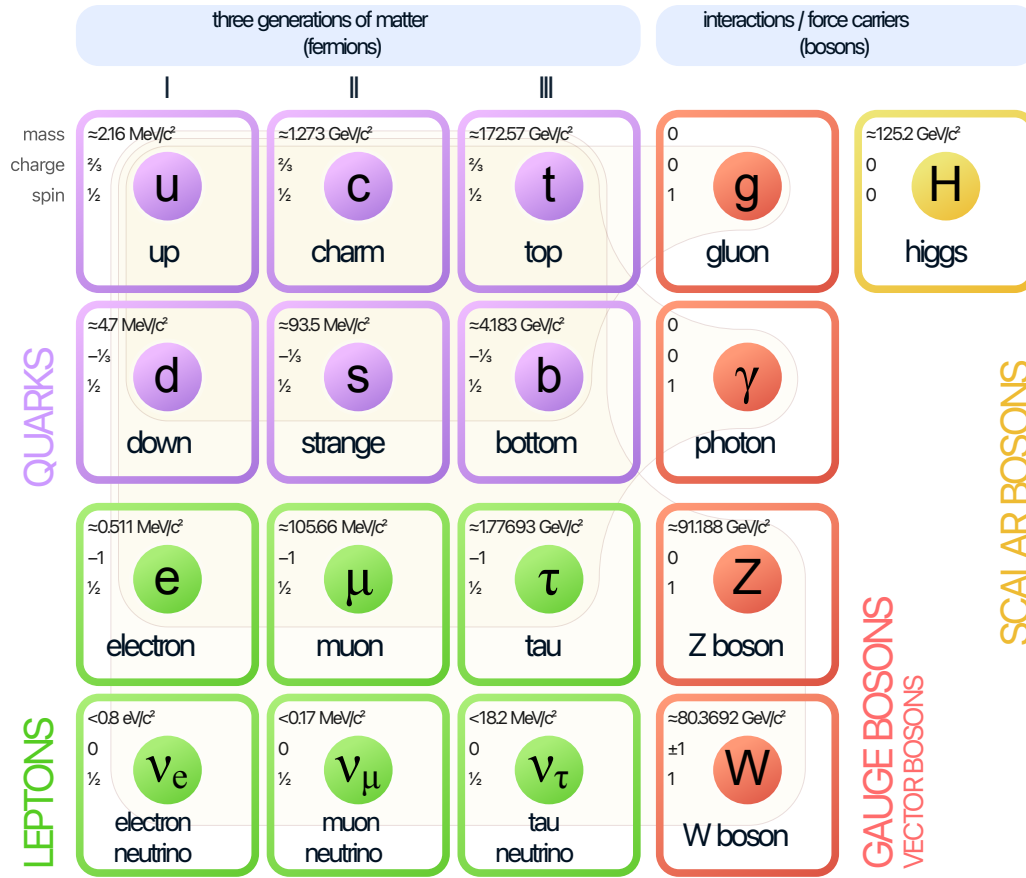


Figure 1: The elementary particles and force carriers of the Standard Model. Adapted from Ref. [1].

1.2 QCD

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1.3 Jets

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1.4 Parton Distribution Functions

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Chapter 2 High Energy Nuclear Collisions

2.1 accelerator considerations for nuclear species

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Chapter 3 Quark Gluon Plasma

3.1 soft probes of the QGP

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Chapter 4 The LHC and ATLAS

4.1 the LHC

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4.2 ATLAS detector

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4.3 ZDC detector

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4.4 ATLAS jet reconstruction

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Chapter 5 $p+A$ collisions

5.1 cold nuclear matter effects

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5.2 jets in $p+A$ collisions

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5.3 color trasnperencey (GPD) in $p+A$ collisions

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Chapter 6 Dependence of geometry estimators on hard scattering kinematics in p+Pb collisions

Just like A+A collisions, hard-process rates in p +A collisions are studied as a function of centrality, defined using global event-level observables sensitive to collision geometry. In the presence of a hard scattering, however, the proton configuration can modify the relation between those observables and the impact parameter, contaminating the above correlation. Measuring the estimators directly as a function of the reconstructed partonic kinematics inverts the usual centrality-selected approach and quantifies this bias.

This chapter presents the first measurement of event geometry estimators on hard-scattering kinematics in p +Pb collisions. Dijets are used as a proxy for the initial-state partonic scattering configuration [2] and their kinematics are correlated with the Pb-going flow forward transverse energy and zero-degree neutral energy.

6.1 Measured Observables

The measured observables are the neutral energy deposited in the Pb-going Zero Degree Calorimeter (ZDC), $E_{\text{ZDC}}^{\text{Pb-side}}$, and the transverse energy measured in the Pb-going Forward Calorimeter (FCal), E_{T}^{Pb} . The ZDC is sensitive primarily to spectator neutrons emitted during nuclear breakup, whereas the FCal transverse energy is an event-activity estimator sensitive to particle production by the collision participants. Both quantities have been used to characterize the geometry of proton–nucleus collisions.

The hard-scattering kinematics are accessed using the two highest-transverse momentum jets in each event. The momentum fraction carried by the parton in the proton is estimated at leading order as

$$x_p = \frac{p_{\text{T},1} e^{y_1^{\text{CM}}} + p_{\text{T},2} e^{y_2^{\text{CM}}}}{\sqrt{s_{\text{NN}}}}, \quad (1)$$

where $p_{\text{T},1}$ and $p_{\text{T},2}$ are the transverse momenta of the leading and subleading jets and

y_1^{CM} and y_2^{CM} are their rapidities in the nucleon–nucleon center-of-mass frame. This estimator reconstructs the Bjorken- x of the scattered proton parton from final-state jet kinematics. Simulation studies show that the reconstructed quantity is, on average, 6–8% lower than the parton-level value in the measurement phase space, with a relative resolution that improves from about 10% at low x_p to 4% at high x_p .

Normalized $E_{\text{ZDC}}^{\text{Pb-side}}$ and E_{T}^{Pb} distributions are measured in intervals of x_p , along with their means, $\langle E_{\text{ZDC}}^{\text{Pb-side}} \rangle$ and $\langle \text{FCal} \sum E_{\text{T}}^{\text{Pb-side}} \rangle$. The distributions expose changes in shape, while the means provide compact measures of the x_p dependence. The correlation between $E_{\text{ZDC}}^{\text{Pb-side}}$ and E_{T}^{Pb} is also studied in each x_p interval to connect the spectator-neutron and participant-activity observables.

6.2 Dataset and Event Selection

6.2.1 2016 p +Pb Data

The p +Pb data at $\sqrt{s_{\text{NN}}} = 8.16$ TeV used for this analysis were delivered by the LHC to ATLAS in November and December of 2016. In p +Pb collisions at 8.16 TeV the colliding beams have different energies per nucleon (6.5 TeV protons colliding with $(Z/A) \times 6.5 \text{ TeV} \simeq 2.6 \text{ TeV}$ per nucleon in a lead nucleus), resulting in a collision system with $\sqrt{s_{\text{NN}}} = 8.16$ TeV per nucleon pair. Under these conditions, the center-of-mass frame is shifted in rapidity, y , by $\Delta y = 0.465$ in the direction of the proton beam. Data were collected in two different periods, characterized by opposite beam orientations. Due to technical issues with the ZDC instrumentation (see Section 6.2.4), only **Period B** was used for this measurement.

The runs in this period were characterized by a total integrated luminosity of 55.9 nb^{-1} . Beam 1 consisted of protons and Beam 2 of Pb ions. The protons traveled toward negative rapidities in the “A→C” direction, corresponding to clockwise motion in the accelerator according to ATLAS conventions. This is denoted as the p +Pb configuration. All data sets used in this analysis, listed in Table 1, were processed using

2016 $\sqrt{s_{\text{NN}}} = 8.16$ TeV Data Samples	Luminosity [nb ⁻¹]
data16_hip8TeV.00313063.physics_Main.merge.AOD.r11416_p3868	0.021
data16_hip8TeV.00313067.physics_Main.merge.AOD.r11416_p3868	0.486
data16_hip8TeV.00313100.physics_Main.merge.AOD.r11416_p3868	8.787
data16_hip8TeV.00313107.physics_Main.merge.AOD.r11416_p3868	11.053
data16_hip8TeV.00313136.physics_Main.merge.AOD.r11416_p3868	9.602
data16_hip8TeV.00313187.physics_Main.merge.AOD.r11416_p3868	3.030
data16_hip8TeV.00313259.physics_Main.merge.AOD.r11416_p3868	4.659
data16_hip8TeV.00313285.physics_Main.merge.AOD.r11416_p3868	4.272
data16_hip8TeV.00313295.physics_Main.merge.AOD.r11416_p3868	9.923
data16_hip8TeV.00313333.physics_Main.merge.AOD.r11416_p3868	3.791
data16_hip8TeV.00313435.physics_Main.merge.AOD.r11416_p3868	0.301

Table 1: Data samples from $\sqrt{s_{\text{NN}}} = 8.16$ TeV p +Pb collisions collected during the 2016 Heavy Ion Run. The luminosity collected for each run is reported. All the data were processed using ATHENA Release 21.

6.2.2 Trigger Configuration

6.2.3 Monte Carlo Samples

In order to evaluate the performance of jet reconstruction and the jet response within ATLAS, dijet Monte Carlo samples generated using PYTHIA8 [3] with the A14 tune and the NNPDF23LO PDFs [4] were produced. Dijet events are filtered using the JZ*R04 filters, which select events containing an anti- k_t $\tilde{R} = 0.4$ jet in specific truth p_T ranges, referred to using the prefix ‘JZ’ and an integer ranging from 1–5 in this analysis. The detector response was simulated using the full ATLAS detector simulation based on the Geant4 toolkit [5]. For p +Pb collisions, Pythia8 truth events are overlaid with real minimum-bias p +Pb data. For each event, the Pythia generation and subsequent simulations were run with conditions matching the collected data, allowing for the simulation to account for underlying event effects in the jet reconstruction. Please note that, in this MC sample, the event activity is originated mainly from the overlay contribution, and therefore has no truth associated.

Events are required to satisfy detector and data-quality requirements, contain at least one reconstructed primary vertex, and contain two reconstructed jets. Fully efficient central and forward single-jet triggers with different transverse-momentum thresholds provide coverage over the analysis phase space. Jets are reconstructed from calorimeter towers with the anti- k_t algorithm, using E-scheme recombination and radius parameter $R = 0.4$. The leading and subleading jets are required to satisfy $p_{T,1} > 40$ GeV and $p_{T,2} > 30$ GeV, respectively, with $-2.8 < \eta_{1,2} < 4.5$. The asymmetric pseudorapidity selection prevents the jets from biasing the Pb-going FCal estimator.

Events with either selected jet near the disabled hadronic endcap calorimeter sector are removed. The excluded region is enlarged by 0.4 in both pseudorapidity and azimuth, and the corresponding efficiency is included in the detector correction. The analysis considers the 0–90% event-activity interval, suppressing ultra-peripheral-collision contamination. In-time pileup is rejected by requiring that no non-primary vertex has more than six associated tracks. A ZDC pre-sample requirement rejects events with elevated baselines from preceding interactions, thereby removing residual out-of-time pileup that can distort the FCal energy measurement.

Detector effects are evaluated with PYTHIA dijet events overlaid with minimum-bias p +Pb data during detector simulation. The simulated FCal distribution is reweighted at the event level to match that of the dijet data.

6.2.4 Missing ZDC Module on Side C

During the 2016 p +Pb run, the ZDC side C module 0 (EM) was removed and replaced with the Arm2 calorimeter of the LHCf experiment [6]. Figure 2 displays the transverse acceptance of both the ATLAS ZDC and LHCf. The ATLAS ZDC EM Module is the first of 4 sequential calorimeter modules (counting from the perspective of looking at the ZDCs from the ATLAS interaction point); therefore, the LHCf calorimeter would see forward neutral products before they reach the side C ZDC. The LHCf detector was put

in place to measure the forward neutral products of the proton going side in the $p+\text{Pb}$ orientation; however, in the $\text{Pb}+p$ orientation the LHCf detector was retracted into the “garage” position to avoid radiation damage [6]. The “garage” position indicates an upwards vertical shift of 112 mm from the nominal position shown in Figure 2. This shift places the full LHCf Arm2 calorimeter outside of the full acceptance of the ATLAS ZDC. Still, the side C ZDC suffers from only having roughly 75 % of the material budget of side A. This effectively reduces the interaction length of the side C ZDC from 4.6 to 3.4, which in turn reduces the neutrons expected to shower in the ZDC from 99% to 96.6%. For this reason, in this analysis only Period B ($p+\text{Pb}$), is used to obtain the final results.

In the $p+\text{Pb}$ orientation, the LHCf was in running position in runs 313063 and 313435, while for all other runs it was in “garage” position. Due to the asymmetric design of the LHCf module in the transverse plane, the effect on the acceptance of the ZDC in the proton-going direction during these runs is nontrivial. Any measurement in this thesis using the proton-going ZDC excludes these runs.

6.3 Analysis Procedure

The Pb-going FCal transverse energy is constructed by summing the transverse projection of the electromagnetic-scale tower energies in the hemisphere facing the Pb beam. This definition is consistent with previous ATLAS $p+\text{Pb}$ centrality measurements. In typical events containing a hard scattering, the energy observed in the Pb-going ZDC is dominated by spectator neutrons evaporating from the struck nucleus.

The ZDC energy scale is determined run by run from the response to a single spectator neutron. The low-energy spectrum contains distinguishable neutron peaks, and a simultaneous fit to the first three peaks and a background term validates the calibration. The fitted single-neutron peak is constrained to the nominal per-nucleon beam energy. The detector response below 1 TeV is limited by detector noise and low-energy backgrounds; this region is treated as a dedicated zero-energy bin. A residual

in-time pileup contribution to the $E_{\text{ZDC}}^{\text{Pb-side}}$ distribution is removed with a data-driven template method.

Finite jet-resolution effects cause migration in x_p . These effects are corrected with a two-dimensional iterative Bayesian unfolding procedure using RooUnfold. The response matrix contains x_p and, in turn, either $E_{\text{ZDC}}^{\text{Pb-side}}$ or E_{T}^{Pb} . Since the available simulation does not model nuclear breakup in the ZDC or the full FCal response for the overlaid minimum-bias event, the calorimeter observable is propagated through the unfolding with diagonal migration matrices. This retains the measured correlation of calorimeter energy with x_p while correcting the jet-related kinematic migration.

The response matrix is reweighted as a function of reconstructed x_p to match the data spectrum. For the ZDC measurement, pileup-subtracted data distributions are sampled to assign a ZDC energy to each simulated dijet event. The procedure is repeated with independent random seeds, and closure tests verify that the unfolding reproduces the particle-level simulation. Two iterations are used, balancing the convergence of the unfolded result against the growth of statistical uncertainty. Statistical uncertainties are evaluated by bootstrap resampling of both the data and response matrix.

6.4 Systematic Uncertainties

Systematic uncertainties arise from the jet energy scale (JES), jet energy resolution (JER), unfolding prior, disabled HEC sector, and the ZDC energy calibration. Except for the HEC and ZDC-scale variations, the full analysis is repeated with a response matrix modified according to each systematic variation. The difference relative to the nominal result defines the corresponding uncertainty.

The JES and JER variations quantify the mean and width of the $p_{\text{T}}^{\text{reco}}/p_{\text{T}}^{\text{gen}}$ response for matched jets. The HEC uncertainty is evaluated by enlarging the jet-exclusion region by 0.1 in both pseudorapidity and azimuth. To test the sensitivity to the unfolding prior, the response matrices are constructed without the nominal event-level reweighting and

the full unfolding is repeated.

The ZDC-specific uncertainty includes the stability of the single-neutron peak, the residual energy-scale variation, and the non-linear energy correction. The FCal stability across the data-taking period is negligible relative to the other sources. The independent variations are added in quadrature. For ratios between x_p intervals, JES, JER, HEC, and ZDC uncertainties are treated as fully correlated, while the unfolding-prior contribution is treated as uncorrelated.

For the normalized distributions, the $E_{\text{ZDC}}^{\text{Pb-side}}$ uncertainty is typically dominated by the ZDC energy-stability term and is below 7% over most of the measured range. The E_{T}^{Pb} distribution uncertainty is primarily driven by JES, JER, and the unfolding prior and is generally below 10%. The systematic uncertainties on $\langle E_{\text{ZDC}}^{\text{Pb-side}} \rangle$ and $\langle \text{FCal} \sum E_{\text{T}}^{\text{Pb-side}} \rangle$ are below 2% in all but the lowest x_p interval.

6.5 Results

The normalized E_{T}^{Pb} and $E_{\text{ZDC}}^{\text{Pb-side}}$ distributions change systematically with x_p . Relative to a low- x_p selection, high- x_p dijet events exhibit an E_{T}^{Pb} distribution shifted toward lower energies. This is a direct confirmation of the event-activity bias previously observed in centrality-dependent jet measurements. The $E_{\text{ZDC}}^{\text{Pb-side}}$ distribution shows a qualitatively similar trend: at high x_p , a larger fraction of events contains fewer forward neutrons. The effect in the ZDC is, however, substantially weaker than that in the FCal.

The mean values $\langle E_{\text{ZDC}}^{\text{Pb-side}} \rangle$ and $\langle \text{FCal} \sum E_{\text{T}}^{\text{Pb-side}} \rangle$ both decrease with increasing x_p , most visibly for $x_p \gtrsim 0.02$, where earlier ATLAS measurements found evidence for small proton-size configurations. Across this range, $\langle E_{\text{ZDC}}^{\text{Pb-side}} \rangle$ decreases by up to about 5%, corresponding on average to two fewer beam-energy neutrons, whereas $\langle \text{FCal} \sum E_{\text{T}}^{\text{Pb-side}} \rangle$ decreases by up to 40%. When the two means are normalized to their maximum values, the FCal estimator is found to be approximately six times more sensitive to hard-process kinematics than the ZDC estimator.

The relation between E_T^{Pb} and $E_{\text{ZDC}}^{\text{Pb-side}}$ is also x_p dependent. For ZDC energies below approximately 150 TeV, the average FCal transverse energy is consistent with a linear dependence on the ZDC energy in low-, middle-, and high- x_p selections. The slope decreases progressively from low to high x_p , and the Pearson correlation coefficient changes by approximately 3% across the measured range. At higher ZDC energies, deviations from the linear trend are consistent with larger fluctuations in nuclear breakup for events with the greatest event activity.

These results show that both common geometry estimators retain sensitivity to the proton configuration associated with a hard scattering. The ZDC observable is more robust than Pb-going FCal transverse energy, but is not independent of the hard-scattering kinematics. This observation provides experimental input for simultaneous descriptions of event activity, nuclear breakup, and color fluctuations in $p+\text{Pb}$ collisions.

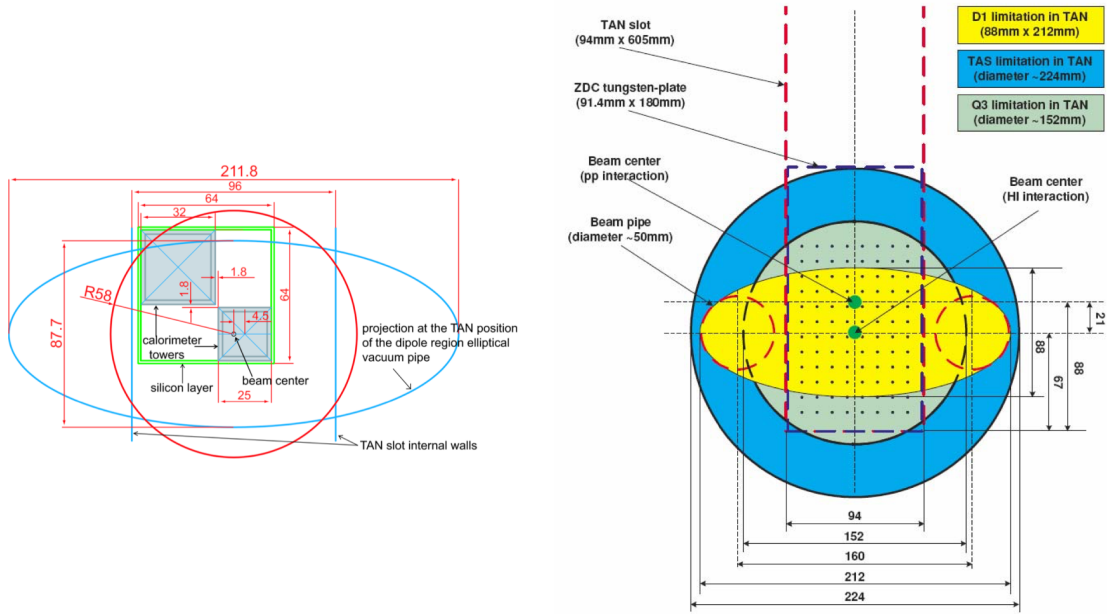


Figure 2: Detector Geometry for LHCf Arm2 [7] (left) and the ATLAS ZDC [8] (right). View is from the perspective of the ATLAS interaction point. In both figures the distances are given in mm. The blue oval in the left image corresponds to the boundary of the yellow oval in the right image. The active area of LHCf is represented by the filled blue squares. The active area of the ZDC is given by the area bounded by the purple dashed line.

Chapter 7 jet quenching in small systems

7.1 theoetical preductions

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7.2 measurments from other collaberations

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Chapter 8 Observation of centrality-dependent dijet transverse momentum imbalance in O+O and Ne+Ne collisions

8.1 observable

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8.2 dataset and event selection

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8.3 analysis

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8.4 uncertainties

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8.5 results

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Chapter 9 Summary and Conclusions

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